

Lab_0: Building Hierarchy and Behavioral Simulation

Introduction

In this lab, you will write a complete RTL description for the modules *MY_AND2* and *MY_OR2* and build the circuit shown below (**Figure 1**) in a structural Verilog description of the top-level module *AND_OR*. Then you will write a Verilog testbench for the *AND_OR* module completed previously. Another important goal of this experiment is to learn how to submit your results.

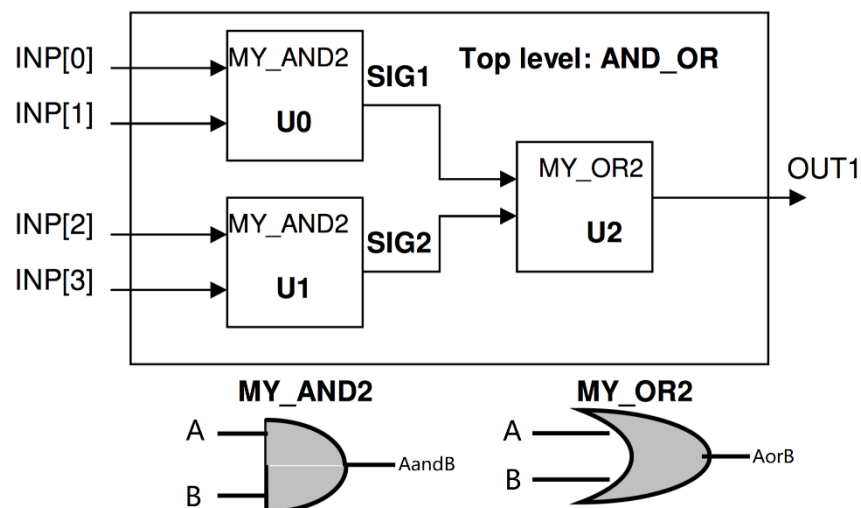


Figure 1 *AND_OR* Structure and Underlying Components

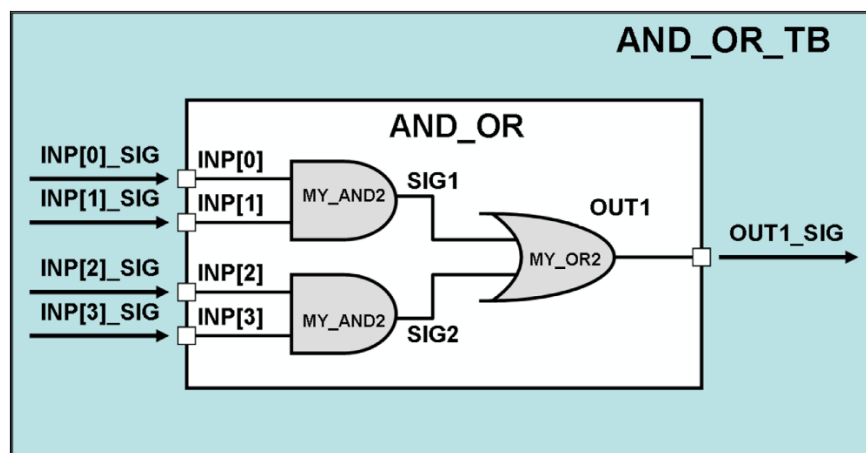


Figure 2 Schematic for Upper-level Testbench with *AND_OR*

Software

You don't need any professional software for this experiment. An editor that supports HDL syntax highlighting (such as Visual Studio code or notepad ++) is enough to write the perfect code. Of course you can't test the code, but I'm sure something as simple as that won't get you wrong.

Later experiments will use Xilinx Vivado. You can download the [installation package](#) on Xilinx's official website and get its evaluation license file.

Key points

1. **The circuit diagram described in your Verilog code must be exactly the same as Figure 1. It is incorrect to write all logic at top-level!**
 - The name of the top-level module should be correct (refer to **Figure 1**).
 - The name of the instances should be correct (refer to **Figure 1**).
 - The name of I/O ports in the top-level should be correct (refer to **Figure 1**).
 - The name of the wires should be correct (refer to **Figure 1**).

Code format

1. Keywords (except data type) are **all lowercase**.
2. Module name is the same as the file name.
3. Naming must be **meaningful** (except for the prescribed name). There are no restrictions on naming, and you can reasonably use uppercase letters and underscores to make your name easier to read and understand.
4. Use **four Spaces** or **a Tab** to indent your code. Reasonable indentation makes your code easier to read and understand.

Submit

1. Compress the HDL code and **name the compressed package as follow the format:**
Lab0_[YourNameInEnglish]_[YourStudentNumber].zip

For example:

Lab0_ChenFupeng_12345678.zip

2. **File Organization Schema in Package:**

```
Lab0_[YourNameInEnglish]_[YourStudentNumber].zip
├──MY_AND2.v      port: MY_AND2_A, MY_AND2_B, MY_AND2_C
├──MY_OR2.v       port: MY_OR2_A, MY_OR2_B, MY_OR2_C
├──AND_OR.v       port: INP, OUT1
└──AND_OR_tb.v
```

3. Submit the compressed package.

4. This course uses a combination of automatic grading and manual grading. If you have any questions about your score, please complain to the teaching assistant in time.

Deadline: 2021-9-27 17:30

- Submit on time, get all scores.
- Submission time does not exceed 24 hours of the deadline, get half of the score.
- Submitted more than 24 hours from the deadline, get no score.

Any Question?

Any questions on course or labs can be proposed in the Discussing Forum on BB.

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